Warmup/Attendance: [Week 5: Wednesday 9/23/2026]

Full Name: SID:

Work with the people around you to try to solve these problems, which we'll discuss together once lecture begins. These are not graded for correctness, but you will turn this sheet in for attendance at the end of class.

**1.** If I roll a fair **six**-sided die, what is the chance that I will roll a multiple of 3 OR a multiple of 4?

**2.** We have two events $A$ and $B$ in an outcome space $\Omega$. If $P(A) = 0.7$ and $P(B) = 0.5$, can $A$ and $B$ be mutually exclusive? Draw a Venn diagram and use it to say what is the *largest* that the intersection of $A$ and $B$ could possibly be. Also, what is the *smallest* this value can be. That is, find upper and lower bounds on the value of $P(A \cap B)$.

**3.** There are 365 (or 366) days in a year. Everyone has a birthday (e.g. August 2 or June 14). We have roughly 80 people in this class, so clearly we can't cover the whole calendar year, but it is still *possible* that two people in class have the same birthday.

1. Try to guess, as a percentage, the chance that at least two people somewhere in this class have the same birthday.
2. How many people would you need in a class before the chance that two people have the same birthday is more than 50%? How would you even begin to think about this chance?